

An Authentication Scheme for VQ-Compressed Images

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Abstract—Some data hiding schemes for VQ-compressed images are based on index modifications. These may cause many distortions between host images and watermarked images, so they are not suitable for authentication of VQ-compressed images. In this paper, we propose a new image authentication scheme for VQ-compressed images. This scheme utilizes an information hiding method based on covering codes. It only modifies few indices slightly to conceal authentication information. Moreover, the proposed scheme is quite simple and secret, and can detect any modifications of VQ-compressed images. Theoretical analysis and experimental results demonstrate the effectiveness of this scheme.

Keywords—image authentication; digital watermarking; vector quantization; covering code

I. INTRODUCTION

Digital representation of multimedia content offers various advantages, such as easy and wide distribution of multiple and perfect replications of the original content [1]. At the same time, with the availability of powerful software, it is also very easy even for inexperienced users to manipulate digital content without causing noticeable traces. Therefore authenticity and integrity of the multimedia data are of utmost important and vital whenever decision must be made concerning origin of an image and its content. Watermarking is one of the methods that have been employed to combat the security issues related to digital media piracy and manipulation.

In order to speed up the transmission process on the Internet, where the limited bandwidth has always been a problem, lossy image compression techniques such as vector-quantization-based (VQ-based) [2] methods have advantages in terms of simple implementation and high compression ratio, and have been applied in many applications. In VQ, the image needs to be divided into non-overlapping blocks of $k \times k$ pixels, which are then transformed into k^2 -dimensional vectors before executing the VQ encoding procedure. It is also necessary to design a set of representative image blocks beforehand to form a codebook $Y = \{y_1, y_2, \dots, y_{N-1}\}$, where each element y_i in Y is regarded as the i -th k^2 -dimensional codeword.

On the encoder side, a desired mapping function Q is designed such that the Euclidean distance between the input vector S_j and the mapped codeword $Q(S_j) = y_x$ is the

shortest. The Euclidean distance, defined in (1), is frequently used to measure the similarity.

$$ED(S_j, y_x) = \sqrt{\sum_{i=1}^{k \times k} (S_{ji} - y_{xi})^2}, \text{ for } 0 \leq x \leq N-1. \quad (1)$$

On the decoder side, the transmitted data x is used as the index to extract the corresponding codeword from the same codebook Y so that the input vector S_j can be approximated.

The key point to the design of a perfect VQ scheme is to generate a perfect codebook from the training images. The LBG algorithm, proposed by Linde, Buzo and Gray in 1980 [3], gives a good answer and is probably the most famous codebook design algorithm.

In 1999, Lin and Wang proposed an LSB-like method [4] for VQ-compressed images in which each index code holds one secret bit. This method utilizes the pairwise nearest clustering manner to create many pairs of codewords that are extremely similar to each other. One of the two codewords in the same pair is responsible for implying for the secret “0” embedding, and the other codeword is responsible for implying for the secret “1” embedding. Then, Lu et al. [5] extended it to a t -bit/index embedding method. That is to say, 2^t codewords that are as similar as possible are arranged in the same cluster instead of a pair. In 2002, Jo and Kim also designed a revised version [6] of Lin and Wang’s method in order to raise the quality of embedded images. They mainly extract the singular codeword that might burden the quality with large degradation from the codebook and collect them in a cluster that is not for embedding. Further, some adaptive embedding methods, such as [7-9], that can deliver more secret data into VQ-coded images, have been introduced.

Whether codewords in the codebook are paired/clustered, or the fixed/adaptive embedding is exploited, the schemes mentioned above do have one thing in common: when secret data embedded into indices, many of the selected codewords were not the most similar codewords in codebook. This may cause perceptual degradation of images. So they are not suitable for image authentication.

In 2003, Galand and Kabatiansky [10] proposed method of information hiding based on covering codes. It allows embedding up to $\lfloor \log_2(n+1) \rfloor$ bits in coverwords of n bits, changing at most one bit. In this paper, we shall propose a low degradation authentication scheme for VQ-compressed images. This scheme utilizes Galand and Kabatiansky’s method to modify few indices slightly to conceal authentication information. The rest of this paper is

organized as follows. The information hiding method is introduced in Section II. In Section III, we shall present the authentication scheme for VQ-compressed images. Experimental results are illustrated in Section IV. Finally, conclusions are given.

II. INFORMATION HIDING METHOD

This information hiding scheme can embed up to m bits in coverwords of $n = 2^m - 1$ bits, changing at most one bit. Let $c = (c_1, \dots, c_n)^T$ be the binary words which all bits are equally valuable, $d(x, y) = |\{i : x_i \neq y_i\}|$ be the Hamming distance between two words $x = (x_1, \dots, x_n)^T$ and $y = (y_1, \dots, y_n)^T$. The information hiding method can be considered as the following model. The data $b = (b_1, \dots, b_m)^T$ are needed to embed in binary coverword c in an invisible way, i.e. the resulting stegoword c' should look like c . This last property shall mean for us that the Hamming distance between c' and c is 1, namely $d(c', c) = 1$. The embedding mapping can be donated as $E(c, b) = c'$. This means it is possible to extract the embedded data b from stegoword c' , i.e. there exists an extracting function: $D(c') = b$. The principle of information hiding method is described in detail in the following paragraphs.

Let H be $m \times n$ parity check matrix of a Hamming code of length $n = 2^m - 1$. We define the m -dimensional vector d as

$$d = Hc = h_1c_1 \oplus h_2c_2 \oplus \dots \oplus h_nc_n. \quad (2)$$

Here, $h_i, 1 \leq i \leq n$ is column vector of matrix H , and $\oplus$ is the XOR operation.

If b is the binary words to be embedded, define the m -dimensional vector e as

$$e = d \oplus b. \quad (3)$$

If $e = 0$, we know that $b = d$, and do not modify the vector c , i.e. $c' = c$; otherwise, we can find a column vector h_i from matrix H where $h_i = e$. We modify the vector c as

$$c' = (c_1, \dots, c_i \oplus 1, \dots, c_n)^T. \quad (4)$$

In the extraction phrase, we can get the embedded data b from codeword c' by Hc' .

If $e = 0$, we know that $b = d$ and $c' = c$, so $Hc' = Hc = d = b$; otherwise, there is a column vector h_i from matrix H where $h_i = e$, so

$$\begin{aligned} Hc' &= h_1c_1 \oplus \dots \oplus h_i c_i \oplus h_i \oplus \dots \oplus h_n c_n \\ &= h_1c_1 \oplus \dots \oplus h_i c_i \oplus \dots \oplus h_n c_n \oplus h_i \\ &= d \oplus e \\ &= b. \end{aligned} \quad (5)$$

From the principle of this method, we can know that if attackers get the matrix H , they can forge another binary words c'' , and let $Hc'' = Hc$. This is not secure for image

authentication, so we need to scramble c before embedding data b into it.

III. THE PROPOSED AUTHENTICATION SCHEME

In this scheme, we first need a VQ codebook of size N . The codebook needs to be ranged so that indices can embed information. We define the codebook as $Y = \{y_0, y_1, \dots, y_{N-1}\}$, which is arranged such that y_{2h} and y_{2h+1} are as similar as possible, where $h = 0, 1, \dots, (N-2)/2$. The codebook is arranged by the following steps: Initially, all the codewords are put into an unclassified list L . Next, the first codeword l_0 is taken to start the pairing operation. From this list, the best codeword l_i to l_0 is chosen according to the criterion of the Euclidean distance, ED , shown in (1). These two codewords, l_0 and l_i , are then drawn out of L and put into codebook Y as a pair. After that, we select another codeword with the smallest index from L to start the above pairing operation. When the list is empty, we obtain $N/2$ pairs. Due to the similarities in the pair of codewords y_{2h} and y_{2h+1} , they can replace each other, and this behavior will be rarely perceived. So we can use LSBs of indices to hide the authentication data.

Next, the original image is divided into non-overlapping blocks and encoded using the arranged codebook. We need to generate authentication data from the VQ-compressed image. The function of authentication data in this scheme is like digital signature. It should be sensitive to image tampering. In this scheme, we generate the authentication data from indices by calculating the hash from all indices. Because few indices need to be modified to embed authentication data, we extract $\lceil \log_2 N \rceil - 1$ MSBs of indices which may need to be modified and all bits of other indices, and convert them into a binary bitstream D . Then binary bitstream D are hashed using a secure key-dependent hash. The hash of 128 bits are computed as

$$A = \text{hash}(D, \text{key}) \quad (6)$$

where key is the secure key. The authentication data can be donated as $A = (a_0, a_1, \dots, a_{127})$ where $a_j = \{0, 1\}$ and $0 \leq j \leq 127$.

Finally, the authentication data need to be hidden into the VQ-compressed image using the information hiding method introduced in Section II. The data-embedding phase and the extraction and verification phase are described in greater detail in the following paragraphs.

A. Data-embedding Phase

The proposed data-embedding phase consists of the following steps.

Step 1. The original image is divided into non-overlapping blocks and encoded using the arranged codebook.

Step 2. Scramble the indices, then generate authentication data from VQ-compressed image.

Step 3. The indices are scanned in a raster-scanning order. The LSBs of the first n indices are extracted as a coverword to hide the first m bits authentication data.

From the principle of information hiding method, only one LSB of n indices needs to be modified, i.e. if the LSB of index i needs to be modified and i is even, the index is modified to $i+1$; if i is odd, the index is modified to $i-1$.

Step 4. The LSBs of the next n indices are extracted as a coverword to hide the next m bits authentication data.

Step 5. Step 4 is repeated until all authentication data are hidden into the indices. Then scramble the indices in reverse.

The stego-image is thus generated, ready for transmission to the receivers. Receivers can extract the secret data from the indices using the proposed scheme to verify the integrity of compressed image.

B. Extraction and Verification Phase

The messages that come across to the receiver are indices. The authentication data can be extracted by reversing the embedding phase described in the previous subsection. Extraction and verification Phase consists of the following steps.

Step 1. Scramble the indices, and then the indices are scanned in a raster-scanning order. The LSBs of the first n indices are extracted as a coverword to extract the first m bits authentication data.

Step 2. The LSBs of the next n indices are extracted as a coverword to extract the next m bits authentication data.

Step 3. Step 2 is repeated until all authentication data are extracted from the indices.

Step 4. Calculate the hash of $\lceil \log_2 n \rceil - 1$ MSBs of all indices, verify the integrity of the compressed image.

If the image is authentic, indices are used to extract the corresponding codeword from the same codebook Y so that the original image can be approximated.

IV. EXPERIMENTAL RESULTS

In this section, we shall show our experimental results to confirm our proposed scheme, and some discussions are made to demonstrate the effectiveness and the feasibility of our scheme. In our experiments, we used MATLAB 7.0 to implement the methods. The effectiveness measurement concerns the visual quality of the compressed image and the watermarked image. The peak-signal-noise-ratio (PSNR) is used to evaluate the visual quality. The PSNR value is defined as follows.

$$MSE = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (I_{ij} - I'_{ij})^2 \quad (7)$$

$$PSNR = 10 \log_{10} \frac{255^2}{MSE} \quad (8)$$

Here, H and W are the height and width of the images in pixel. Notations I_{ij} and I'_{ij} indicate the pixel value of the image I and I' at coordinate (i, j) . The notation MSE , standing for the mean square error, is used to measure the difference between images I and I' .

We perform simulations on several test images of size 512×512 pixels. Test images are divided into non-overlapping blocks of 4×4 pixels. The codebook of size 512 is obtained by the well-know LBG algorithm [3] from test images. The length of coverword is set to 255, so we can hide 8 bits secret data in every 255 indices by slightly changing at most one index, so we only need to change at most 16 indices of VQ-compressed image to hide the authentication data. Figure 1 shows the VQ-compressed images and watermarked images with PSNRs.

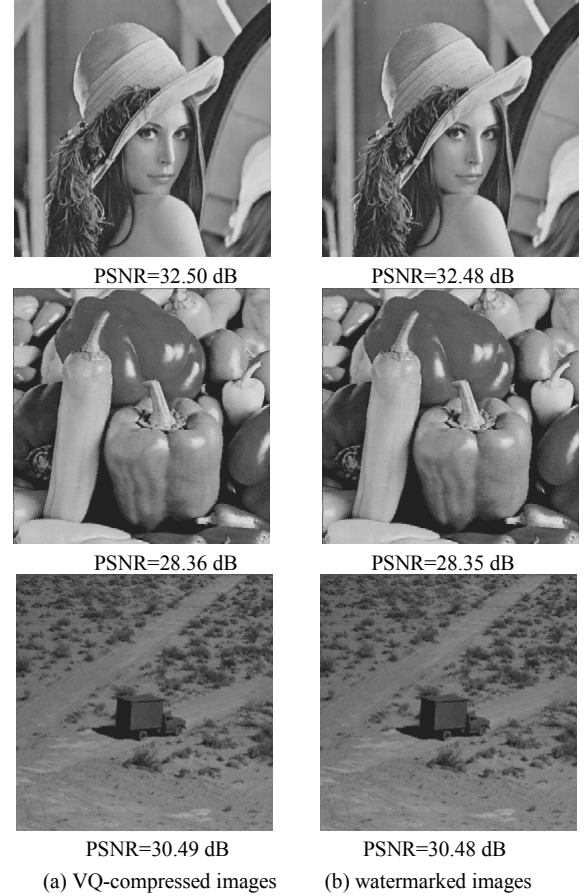


Figure 1. VQ-compressed images and watermarked images

From these results, it can be seen that the proposed scheme only causes little extra distortion. Experimental results also show that the proposed authentication scheme can detect if the key is incorrect, if the image is not watermarked, or if the image is changed in its block indices.

V. CONCLUSION

In this paper, a new image authentication scheme for VQ-compressed image which utilizes an information hiding method based on covering codes is presented. It only modifies few indices slightly to conceal authentication data, so there are only few distortions between host image and its watermarked image. The proposed scheme is sample, secret and efficient. Further work will concentrate on tampering localization capability for VQ-compressed images.

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